Boudrouss Réda | CV

Software Engineer – Static Analysis, Compilers & Formal Methods – Paris

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Languages: French (Native), English (Professional), Arabic (Conversational)

Profile

Software engineer completing an MSc in Computer Science (Software Science & Technology) at Sorbonne University, specialized in **static analysis by abstract interpretation**, **formal program verification** and the implementation of programming languages and runtimes. I ported **MOPSA**, an abstract-interpretation analyzer for C and Python developed at LIP6 (OCaml/C/C++), to WebAssembly, and built an abstract-interpretation-based analyzer in **Rust**. At Hexaglobe I industrialized LLM-agent test production gated by **deterministic verification signals** such as mandatory green tests, with sandboxed orchestration and systematic human review.

Projects

Mopsa-wasm - github.com/rboudrouss/mopsa-emcc — 2026

· Ported **MOPSA**, an **abstract-interpretation** static analyzer for C and Python (OCaml/C/C++, developed at LIP6), to **WebAssembly**, compiling the OCaml runtime and the full native stack (Clang frontend, **Apron**, GMP, MPFR) into a single static Wasm module with Emscripten.

· Engineered a two-stage LLVM/Clang cross-compilation (native TableGen tools first, then a Wasm frontend-only build) to embed a C parser in the browser, and generated a static table of ∼1,400 C FFI primitives to replace dynamic loading.

· Addressed the **soundness** hazard of Wasm's fixed rounding mode (no `fesetround`) by recompiling Apron's numerical domains to use exact GMP rationals.

· Shipped a fully client-side playground and a technical write-up.

Reactant analyzer - github.com/rboudrouss/reactant-analyzer — 2026

· Designed and implemented an **abstract-interpretation**-based static analyzer written in **Rust**, targeting React web applications.

Education

MSc in Computer Science (Software Science & Technology) - Sorbonne University — 2024 – 2026

· Coursework: **static analysis by abstract interpretation** and type systems (course led by Antoine Miné), inference of numerical properties, program specification & **formal verification**, compiler construction, design & implementation of programming languages and runtimes, concurrent & distributed programming.

BSc in Computer Science - Sorbonne University — 2020 – 2024

· Algorithms, computer architecture, operating systems, networks, databases, functional and object-oriented programming, with team projects in C, Java, OCaml, Python and TypeScript.

Experience

Quality Analyst Engineer (apprenticeship) - Hexaglobe (video streaming) — 2025 – 2026

· Industrialized **LLM-agent test production** gated by deterministic **verification signals** (mandatory green tests), with sandboxed planner/implementer/reviewer orchestration in Docker, versioned guardrails and systematic human review, producing ∼720 integration tests in 3 days.

· Designed objective, automated **verification oracles** (VMAF, black-frame/freeze detection, audio analysis) to validate a live video encoder.

· Brought zero-coverage legacy products to ∼**3,000 test cases** across 4 stacks (TypeScript, Python, PHP, JS), and built **GitLab CI/CD** test pipelines with shared components consumed by 4+ repositories.

Technical Lead, then President (2024–25) - Association Play Sorbonne Université — 2023 – 2026

· Designed and operated a virtualized **infrastructure** (Docker servers, CI/CD, networking). As president, led **100+ volunteers** for a cultural event welcoming 10,000+ visitors.

Skills

Languages: OCaml, Rust, C, Python, TypeScript, Java, SQL
Topics: abstract interpretation & static analysis, compilers & toolchains (LLVM, Emscripten), WebAssembly, formal verification, GitLab CI/CD, Docker